

CS 70 — Discussion 2B

Problem 1: Edge Colorings

An edge coloring assigns colors to edges so that any two edges incident to the same vertex have different colors. Equivalently, each color class is a matching (a set of pairwise non-adjacent edges).

Part (a): The complete graph K_4 is 3-edge-colorable

K_4 has 4 vertices, each of degree 3, and 6 edges. Since each vertex has degree 3, at least 3 colors are needed. We show 3 suffice by partitioning the edges into 3 matchings, labeling the vertices 1, 2, 3, 4:

- **Color 1:** $\{1, 2\}$ and $\{3, 4\}$
- **Color 2:** $\{1, 3\}$ and $\{2, 4\}$
- **Color 3:** $\{1, 4\}$ and $\{2, 3\}$

Each pair consists of disjoint edges, so each color is a valid matching. At every vertex the three incident edges receive colors 1, 2, 3 — all distinct. Hence K_4 is 3-edge-colorable. $\square$

Part (b): Edge coloring the 3-dimensional hypercube

The 3-dimensional hypercube has vertices given by all 3-bit binary strings, with an edge between two vertices exactly when they differ in a single bit. Every vertex has degree 3, so at least 3 colors are required.

Color each edge by the bit position in which its two endpoints differ (color 1, 2, or 3). At any vertex, its three edges flip bits 1, 2, and 3 respectively, so they receive three different colors. Therefore **3 colors** suffice, and 3 is the answer.

Part (c): Any graph with maximum degree d is $(2d - 1)$ -edge-colorable

Proof by induction on $m = |E|$.

Base case. $m = 0$: no edges to color; trivially colorable with any palette.

Inductive hypothesis. Assume any graph with m edges and maximum degree at most d can be edge-colored with $2d - 1$ colors.

Inductive step. Let G have $m + 1$ edges and maximum degree d . Remove an edge $e = \{u, v\}$ to obtain G' , which has m edges and maximum degree at most d . By the inductive hypothesis, G' can be edge-colored with $2d - 1$ colors. Now add e back. The edges that constrain the color of e are those incident to u or v :

- At u : at most $d - 1$ other edges, blocking at most $d - 1$ colors.
- At v : at most $d - 1$ other edges, blocking at most $d - 1$ colors.

So at most $(d - 1) + (d - 1) = 2d - 2$ colors are blocked. Since the palette has $2d - 1$ colors, at least one color remains free for e . Coloring e with a free color completes a valid $(2d - 1)$ -edge-coloring of G . $\square$

Part (d): A tree is d -edge-colorable, where d is the maximum degree

Proof by induction on $m = |E|$.

Base case. $m = 0$: no edges; trivially colorable.

Inductive hypothesis. Assume any tree with m edges and maximum degree at most d can be edge-colored with d colors.

Inductive step. Let T be a tree with $m + 1$ edges and maximum degree d . Since T has at least one edge, it has a leaf v (a vertex of degree 1); let $e = \{u, v\}$ be its unique incident edge. Remove e (and the isolated vertex v) to obtain T' , which is still a single tree with m edges and maximum degree at most d . By the inductive hypothesis, T' can be edge-colored with d colors. Now add e back:

- At v : v is a leaf, so it has no other edges — 0 colors blocked.
- At u : at most $d - 1$ other edges, blocking at most $d - 1$ colors.

So at most $d - 1$ colors are blocked, and with d colors available at least one is free for e . This completes a valid d -edge-coloring of T . $\square$

The improvement over part (c) comes from removing a *leaf* edge, so only one endpoint (u) has previously colored edges.

Problem 2: Bipartite $\iff$ No Odd Tours

Claim: A graph G is bipartite if and only if it has no tours (closed walks) of odd length.

Forward direction ($\Rightarrow$)

Assume G is bipartite with parts L and R . Every edge goes between L and R , so each step of any walk switches sides. Starting from a vertex in L , after k steps the walk is in L if k is even and in R if k is odd. A tour returns to its starting vertex, which is on the same side it began; this requires an even number of steps. Therefore every tour has even length, so G has no odd tours. $\square$

Reverse direction ($\Leftarrow$)

Assume G has no odd tours. Without loss of generality assume G is connected (otherwise apply the argument to each component). Fix a vertex s , and for each vertex v let $\text{dist}(v)$ denote the length of a shortest path from s to v . Define:

$$L = \{v : \text{dist}(v) \text{ is even}\}, \quad R = \{v : \text{dist}(v) \text{ is odd}\}.$$

This partitions the vertices. We claim no edge joins two vertices in the same set.

Suppose for contradiction that $\{u, v\}$ is an edge with u, v in the same set, so $\text{dist}(u)$ and $\text{dist}(v)$ have the same parity. Consider the closed walk formed by a shortest path $s \rightarrow u$, the edge $\{u, v\}$, and a shortest path $v \rightarrow s$. Its length is:

$$\text{dist}(u) + 1 + \text{dist}(v).$$

Since $\text{dist}(u)$ and $\text{dist}(v)$ have the same parity, their sum is even, so the total length is odd. This is an odd tour, contradicting our assumption. Therefore every edge joins a vertex in L to a vertex in R , so G is bipartite. $\square$

Problem 3: Modular Arithmetic Equations

Part (a): $9x \equiv 1 \pmod{11}$

We seek the inverse of 9 modulo 11 via the extended Euclidean algorithm on $\text{gcd}(11, 9)$:

$$11 = 1 \cdot 9 + 2$$

$$9 = 4 \cdot 2 + 1$$

$$2 = 2 \cdot 1 + 0$$

The gcd is 1, so an inverse exists. Back-substituting:

$$1 = 9 - 4 \cdot 2$$

$$= 9 - 4(11 - 9)$$

$$= 5 \cdot 9 - 4 \cdot 11$$

Taking this mod 11: $5 \cdot 9 \equiv 1 \pmod{11}$, so $\boxed{x \equiv 5 \pmod{11}}$.

Check: $9 \cdot 5 = 45 = 4 \cdot 11 + 1 \equiv 1 \pmod{11}$. $\checkmark$

Part (b): $10x + 23 \equiv 3 \pmod{31}$

Reduce: $10x \equiv 3 - 23 \equiv -20 \pmod{31}$.

Find the inverse of 10 modulo 31 via $\text{gcd}(31, 10)$:

$$31 = 3 \cdot 10 + 1$$

$$10 = 10 \cdot 1 + 0$$

The gcd is 1. Back-substituting directly from the first line:

$$1 = 31 - 3 \cdot 10 \implies -3 \cdot 10 \equiv 1 \pmod{31},$$

so $10^{-1} \equiv -3 \equiv 28 \pmod{31}$. Multiplying both sides of $10x \equiv -20$ by -3 :

$$x \equiv -3 \cdot (-20) \equiv 60 \equiv 29 \pmod{31}.$$

So $\boxed{x \equiv 29 \pmod{31}}$.

Check: $10 \cdot 29 + 23 = 313 = 10 \cdot 31 + 3 \equiv 3 \pmod{31}$. ✓

Part (c): $3x + 15 \equiv 4 \pmod{21}$

Reduce: $3x \equiv 4 - 15 \equiv -11 \equiv 10 \pmod{21}$.

The equation $ax \equiv b \pmod{m}$ has a solution if and only if $\gcd(a, m) \mid b$. Here $\gcd(3, 21) = 3$, and $3 \nmid 10$. Therefore **no solution exists**.

Part (d): System modulo 7

$$3x + 2y \equiv 0 \pmod{7}, \quad 2x + y \equiv 4 \pmod{7}.$$

From the second equation, $y \equiv 4 - 2x \pmod{7}$. Substitute into the first:

$$3x + 2(4 - 2x) \equiv 0 \pmod{7} \implies 3x + 8 - 4x \equiv 0 \implies -x + 1 \equiv 0 \pmod{7},$$

using $8 \equiv 1 \pmod{7}$. Thus $x \equiv 1 \pmod{7}$. Back-substituting:

$$y \equiv 4 - 2(1) \equiv 2 \pmod{7}.$$

So $\boxed{x \equiv 1, y \equiv 2 \pmod{7}}$.

Check: $3(1) + 2(2) = 7 \equiv 0 \pmod{7}$ ✓ and $2(1) + 2 = 4 \equiv 4 \pmod{7}$ ✓